

AI INVESTMENT SIMULATOR

GREEN CITY

*Portfolio Management - Detailed Teaching Notes***Portals 19-27 | Learn how to manage investments as conditions change**

The learner enters Green City with an existing portfolio. The task is no longer merely to find investments; it is to manage a changing portfolio deliberately.

Green City purpose

Purple City established investing foundations. Blue City developed analysis and asset selection. Green City develops portfolio management: bonds and fixed income, ETFs and alternatives, strategic allocation, diversification, rebalancing, macroeconomics, new information, risk management and crisis response.

How calculations are taught in these notes

- State the formula before using numbers.
- Define the variables and units.
- Substitute the actual numbers into the formula.
- Show intermediate steps rather than jumping to the answer.
- Convert percentages to decimals where necessary.
- Interpret the numerical answer in plain language.
- Explain how the result affects a portfolio decision.

Green City Roadmap

Portal	Topic	Core development	Portfolio application
19	Bonds & Fixed Income	Government/corporate bonds, yields, duration, credit risk and interest-rate sensitivity.	Decide whether fixed income improves the portfolio.
20	ETFs, Commodities & Alternatives	Index funds, sector ETFs, gold, commodities and alternative exposures.	Compare direct ownership with pooled and alternative exposures.
21	Strategic Asset Allocation	Long-term allocation across stocks, businesses, property, forex, bonds, alternatives and cash.	Create a target multi-asset portfolio.
22	Portfolio Diversification	Correlation, concentration, sector, geographic and currency exposure, and liquidity.	Identify hidden concentrations and diversification gaps.
23	Add, Trim or Exit	Rebalancing bands, valuation discipline, thesis deterioration, sell rules and capital rotation.	For every holding choose ADD, HOLD, TRIM, EXIT or REPLACE.
24	Macroeconomics for Investors	GDP, inflation, rates, unemployment, fiscal/monetary policy, yield curves and cycles.	Assess implications across each major portfolio asset class.
25	Earnings, News & New Information	Earnings, guidance, analyst revisions, regulation, rate announcements and property-market information.	Distinguish thesis-changing evidence from market noise.
26	Risk Management & Stress Testing	Volatility, beta, drawdown, Sharpe ratio, correlation, VaR, leverage, liquidity and scenario analysis.	Stress-test the portfolio against recession, inflation, property, currency and company shocks.
27	Green City Crisis Challenge	Integrated market-shock simulation across holdings and asset classes.	Decide what to sell, protect, add to or leave untouched under pressure.

Management mindset

Green City is about managing a system, not picking isolated winners. Every decision should consider what the asset contributes to the total portfolio: return, risk, liquidity, correlation, cash-flow needs and vulnerability to changing economic conditions.

Portal 19 - Bonds & Fixed Income

Learning objective	Portfolio Lab application
Understand government and corporate bonds, yields, duration, credit risk and interest-rate sensitivity.	Decide whether fixed income improves the portfolio.

19.1 What is a bond?

Concept bridge: Buying a bond is economically different from buying a share. A shareholder participates in the residual success of a company, while a bondholder has a contractual claim. This distinction explains why bond analysis concentrates on promised cash flows, maturity, interest rates and the issuer's ability to pay.

Think in cash flows: A plain bond can be understood as a schedule of future cash flows. Before maturity the investor normally receives coupon payments; at maturity the investor normally receives the final coupon plus principal. The market price today is what investors are willing to pay for that stream of future cash flows.

Why this matters in Green City: The learner is no longer asking only whether an investment may rise in price. The learner must ask what role the bond plays beside stocks, property, cash and other holdings: income generation, liquidity, capital stability, diversification or a deliberate interest-rate view.

A bond is a debt security. The investor lends money to an issuer such as a government or company. In return, the issuer normally promises periodic interest payments and repayment of principal at maturity. Unlike a stockholder, a bondholder is primarily a creditor rather than an owner.

Term	Meaning
Face/par value	Amount repaid at maturity, commonly quoted as 100 or a currency amount such as \$1,000.
Coupon rate	Contractual annual interest rate applied to face value.
Coupon payment	Cash interest paid to the bondholder.
Maturity	Date when principal is due.
Market price	Current price at which the bond can be bought or sold.
Yield	Return measure implied by price and cash flows.
Credit risk	Risk that issuer cannot make promised payments.
Duration	Measure of interest-rate sensitivity and weighted timing of cash flows.

19.2 Coupon payment

Annual coupon payment = Face value × Coupon rate

Example 19.1 - Annual coupon

A bond has face value \$1,000 and a 6% annual coupon.

Coupon = Face value × Coupon rate

Workings:

Convert 6% to decimal: 6% = 0.06

Coupon = \$1,000 × 0.06

Coupon = \$60

Answer: \$60 per year.

What the answer means: The investor receives \$60 of contractual coupon interest each year, assuming the issuer pays as promised.

Investment implication: Coupon cash flow can help stabilize portfolio income, but the bond's market price may still change.

19.3 Current yield

Current yield = Annual coupon / Current market price × 100%

Example 19.2 - Current yield

A \$1,000 face-value bond pays a \$60 annual coupon and trades at \$950.

Current yield = Annual coupon / Market price × 100%

Workings:

$$\text{Current yield} = \$60 / \$950 \times 100\%$$

$$\$60 / \$950 = 0.0631579$$

$$0.0631579 \times 100\% = 6.31579\%$$

Answer: Approximately 6.32%.

What the answer means: The investor is receiving annual coupon cash equal to about 6.32% of the price currently paid for the bond.

Investment implication: Current yield ignores capital gain/loss at maturity and time value of money, so it is not the same as yield to maturity.

Transition: Coupon tells you the contractual cash payment. Current yield relates that payment to today's market price. The next step is to consider total return, including price changes.

19.4 Bond price and total holding-period return

Holding-period return = (Coupon income + Ending price - Beginning price) / Beginning price × 100%

Example 19.3 - Bond holding-period return

A bond is bought for \$980, pays \$50 coupon income over the year, and is sold for \$1,005.

HPR = (Coupon + P_1 - P_0) / P_0 × 100%

Workings:

Price change = \$1,005 - \$980 = \$25

Total gain = \$50 coupon + \$25 price gain = \$75

HPR = \$75 / \$980 × 100%

\$75 / \$980 = 0.0765306

0.0765306 × 100% = 7.65306%

Answer: Approximately 7.65%.

What the answer means: The investor earned both coupon income and a capital gain.

Investment implication: For portfolio management, bond return is not only coupon income; changing market yields can create gains or losses.

19.5 Why bond prices move inversely to yields

A useful mental model: Imagine that you own a fixed coupon of \$50 per year. If new bonds of similar risk begin offering much more income for the same purchase price, investors will not normally pay the old full price for your lower-paying bond. Its price must fall until its return becomes more competitive. The reverse logic applies when new market yields fall.

Do not confuse coupon rate with market yield: The coupon rate is fixed by the bond contract for a conventional fixed-rate bond. Market yield changes continuously as price, interest-rate expectations, credit conditions and demand change. Therefore a 5% coupon bond does not necessarily offer a 5% return to a new buyer.

Portfolio connection: This price-yield relationship means a bond allocation can lose market value even when every coupon is paid on time. Interest-rate risk and default risk are separate questions and should be assessed separately.

Existing fixed-rate bonds compete with newly issued bonds. If market interest rates rise, new bonds may offer higher coupons or yields. An older bond paying a lower fixed coupon becomes less attractive unless its price falls. If market rates fall, an older higher-coupon bond can become more attractive, pushing its price up.

Simple intuition

Suppose two equally risky bonds promise \$50 per year. If a newly issued bond can be bought for \$1,000 while an older bond also costs \$1,000 but offers a lower effective return, investors will prefer the better-paying security. Price adjusts until expected returns become more competitive.

19.6 Approximate price sensitivity using modified duration

What duration is doing: Modified duration converts an interest-rate change into an approximate percentage price change. It is therefore a sensitivity measure. A duration of 7 does not mean the bond matures in seven years; in this context it means the bond price is approximately seven times as sensitive, in percentage terms, to a small change in yield.

Reading the sign: The minus sign in the duration approximation is essential. A positive change in yield produces an estimated negative price change, while a negative change in yield produces an estimated positive price change.

Limit of the calculation: Duration is a linear approximation. For large rate moves the actual bond-price curve is curved rather than perfectly straight. More advanced analysis adds convexity, but duration remains a powerful first risk estimate for a beginning portfolio manager.

Approximate % price change $\approx$ -Modified duration $\times$ Change in yield

The negative sign represents the inverse relationship between yield and price. The formula is an approximation for relatively small yield changes.

Example 19.4 - Duration and a rate rise

A bond has modified duration 5.2. Market yield rises by 0.75 percentage points.

Approximate $\% \Delta \text{Price} \approx -\text{Duration} \times \Delta \text{Yield}$

Workings:

Convert 0.75 percentage points to decimal: $0.75\% = 0.0075$

$$\% \Delta \text{Price} \approx -5.2 \times 0.0075$$

$$\% \Delta \text{Price} \approx -0.039$$

$$-0.039 \times 100\% = -3.9\%$$

Answer: Approximately a 3.9% price decline.

What the answer means: The bond's price is estimated to fall about 3.9% for this yield increase, before considering convexity and other effects.

Investment implication: Higher-duration bonds expose the portfolio to greater interest-rate sensitivity.

Example 19.5 - Duration and a rate fall

A bond has modified duration 7.0. Yield falls by 0.40 percentage points.

Approximate $\% \Delta \text{Price} \approx -\text{Duration} \times \Delta \text{Yield}$

Workings:

$$\Delta \text{Yield} = -0.40\% = -0.0040$$

$$\% \Delta \text{Price} \approx -7.0 \times (-0.0040)$$

$$\% \Delta \text{Price} \approx +0.028$$

$$0.028 \times 100\% = 2.8\%$$

Answer: Approximately a 2.8% price increase.

What the answer means: Falling yields can produce capital gains, especially for longer-duration bonds.

Investment implication: Duration can be used to balance income objectives against sensitivity to changing interest rates.

19.7 Credit risk

Interest-rate risk versus credit risk: A government bond can fall because market rates rise even if repayment is not in doubt. A corporate bond can also fall because investors become less confident that the company can service its debt. These two forces may occur together.

Why spreads widen: When investors demand more compensation for corporate risk, the corporate yield can rise relative to the government benchmark. A widening spread therefore often signals greater perceived credit or liquidity risk, while a narrowing spread can signal improving confidence.

Research questions: For a corporate issuer, examine leverage, interest coverage, cash generation, refinancing dates, asset quality and the stability of the business. A high yield should never be treated as free additional return.

Government and corporate bonds do not carry identical default risk. Credit risk reflects the possibility that promised interest or principal is not paid in full or on time. Investors often demand a higher yield for bearing greater credit risk.

Credit spread $\approx$ Corporate bond yield - Comparable government bond yield

Example 19.6 - Credit spread

A five-year corporate bond yields 6.4%. A similar-maturity government bond yields 4.1%.

Credit spread = Corporate yield - Government yield

Workings:

Credit spread = 6.4% - 4.1%

Credit spread = 2.3 percentage points

2.3 percentage points = 230 basis points

Answer: 2.3% or 230 basis points.

What the answer means: The corporate bond offers additional yield compensation relative to the government benchmark.

Investment implication: A wider spread may compensate for risk, but can also be a warning that the market expects financial stress.

19.8 What fixed income can contribute to a portfolio

- Income: coupon payments can create planned cash flow.
- Capital preservation: high-quality short-maturity bonds can be less volatile than equities.
- Diversification: certain bonds may respond differently from stocks during economic stress.
- Liability matching: maturity dates can be aligned with future cash needs.
- Deflation/recession exposure: high-quality bonds may benefit when interest rates decline.
- Liquidity: some government bonds trade in deep markets.
- Risk: long duration, inflation and credit deterioration can still create losses.

Transition: Bonds add a new source of return and risk to the portfolio. Portal 20 broadens the toolkit further by showing how pooled vehicles and alternative assets can change exposure without buying every underlying asset directly.

Portal 20 - ETFs, Commodities & Alternatives

Learning objective	Portfolio Lab application
Understand index funds, sector ETFs, gold, commodities and alternative exposures.	Compare direct ownership with pooled and alternative exposures.

20.1 What an ETF does

Ownership structure: An ETF is a wrapper around an investment strategy or basket. The wrapper can make trading convenient, but it does not remove the risks of the underlying holdings. A technology ETF remains exposed to technology companies; a long-duration bond ETF remains exposed to interest rates.

Why beginners can misread ETFs: Two ETFs can both be called diversified while having very different concentration. One may hold thousands of securities across countries and sectors; another may hold a narrow group of companies in one theme. The name of the fund is therefore only the beginning of the research.

Practical due diligence: Read the fund objective, index methodology, top holdings, sector weights, geographic weights, duration where relevant, expense ratio and trading spread. The portfolio manager should be able to explain what economic exposure is being purchased.

An exchange-traded fund pools investor money into a portfolio that trades on an exchange. An ETF may track a broad index, a bond market, a sector, a commodity, a factor or an actively managed strategy. The important question is not merely 'Is this an ETF?' but 'What exposure does this ETF actually contain?'

ETF feature	What to examine
Underlying holdings	What assets are owned?
Index/mandate	What rules determine inclusion and weighting?
Expense ratio	Annual fund operating cost.
Bid-ask spread	Trading friction.
Tracking difference	How closely returns follow the intended benchmark.
Liquidity	Fund trading volume and liquidity of underlying assets.
Concentration	Top holdings, sector and geographic weights.
Currency exposure	Are foreign assets hedged or unhedged?

20.2 Expense ratio calculation

Approximate annual fund fee = Amount invested × Expense ratio

Example 20.1 - ETF expense ratio

An investor places \$8,000 in an ETF with a 0.25% annual expense ratio.

Annual fee ≈ Investment × Expense ratio

Workings:

Convert 0.25% to decimal: 0.0025

Annual fee ≈ \$8,000 × 0.0025

Annual fee ≈ \$20

Answer: Approximately \$20 per year.

What the answer means: The fee is economically small in one year but compounds over long holding periods.

Investment implication: When two funds provide similar exposure, costs can influence long-run net return.

20.3 Tracking difference

Tracking difference = ETF return - Benchmark return

Example 20.2 - Tracking difference

An index rises 9.2% while the ETF tracking it returns 8.9%.

Tracking difference = ETF return - Index return

Workings:

Tracking difference = 8.9% - 9.2%

Tracking difference = -0.3 percentage points

Answer: -0.30 percentage points.

What the answer means: The ETF underperformed its benchmark by 0.30 percentage points over the period.

Investment implication: Fees, trading costs, taxes, cash drag and imperfect replication can create tracking differences.

20.4 Commodities

Why commodities behave differently: A share can generate earnings and a bond can generate contractual coupons. A physical commodity generally does not create an operating cash flow simply by being held. Its investment return therefore depends much more directly on price movement and on the vehicle used to obtain exposure.

Spot price is not always investor return: Commodity ETFs may hold futures contracts rather than the physical commodity. Futures must be rolled from one contract to another. The shape of the futures curve can therefore help or hurt returns even when the quoted spot commodity price changes little.

Portfolio role: Commodities can provide exposure to inflation, supply shortages or geopolitical disruptions, but they can also be highly volatile. Their inclusion should be justified by a portfolio purpose rather than by recent price performance alone.

Commodities include energy products, metals and agricultural goods. They do not produce earnings in the same way as a business. Returns depend heavily on price changes, supply-demand conditions and the structure used to obtain exposure.

Commodity driver	Example
Supply	Oil production cuts, mine closures, crop failures.
Demand	Industrial growth, transportation demand, construction activity.
Inventories	Low inventories can make markets more sensitive to shocks.
Weather	Agriculture and energy demand can be weather-sensitive.
Geopolitics	Sanctions, wars, shipping disruptions.
Currency	Many commodities are priced globally in U.S. dollars.
Futures structure	Contango/backwardation can affect fund returns.

20.5 Commodity return

Price return = (Ending price - Beginning price) / Beginning price × 100%

Example 20.3 - Gold price return

Gold rises from \$2,000 per ounce to \$2,140 per ounce.

Price return = $(P_1 - P_0) / P_0 \times 100\%$

Workings:

Price change = \$2,140 - \$2,000 = \$140

Return = \$140 / \$2,000 × 100%

\$140 / \$2,000 = 0.07

0.07 × 100% = 7.0%

Answer: 7.0% price return.

What the answer means: The value of the gold exposure increased by 7%, ignoring costs and the specific investment vehicle.

Investment implication: Commodity exposure may diversify inflation or geopolitical risk, but price volatility can be substantial.

20.6 Direct ownership vs pooled exposure

Exposure	Direct ownership	ETF/pooled vehicle
Stocks	Own individual company shares.	Own a basket through a fund.

Bonds	Own specific issuer/maturity.	Diversify across many bonds.
Gold	Physical bullion may require storage.	Gold-linked fund is easier to trade.
Property	Direct property involves transaction/management costs.	REIT ETF offers liquid diversified exposure.
Commodities	Physical ownership may be impractical.	Funds may use futures or related instruments.

Transition: Portal 20 expands the menu of investable exposures. Portal 21 asks a harder question: how much of each exposure should the portfolio hold over the long run?

Portal 21 - Strategic Asset Allocation

Learning objective	Portfolio Lab application
Understand long-term allocation across stocks, businesses, property, forex, bonds, alternatives and cash.	Create a target multi-asset portfolio.

21.1 Allocation is the portfolio's architecture

From security selection to portfolio design: Blue City asks whether an individual opportunity is attractive. Green City adds a second question: even if the opportunity is attractive, how much of the total portfolio should be committed to it? An excellent asset can still create a poor portfolio if its weight is excessive.

Weights create discipline: Expressing holdings as percentages makes risk visible. A \$4,000 holding may seem large or small in isolation; in a \$5,000 portfolio it is dominant, while in a \$100,000 portfolio it is modest. Portfolio weights put every position on the same scale.

Constraints come first: Target allocation should reflect liquidity needs, time horizon, loss capacity and investment objectives. Expected return is important, but a portfolio that cannot meet the investor's cash needs or survive a plausible drawdown is poorly designed.

Strategic asset allocation sets long-term target weights based on the investor's objective, horizon, risk capacity, liquidity needs and beliefs about asset behavior. It is not a prediction of which asset will win next month.

Asset weight = Asset value / Total portfolio value × 100%

Example 21.1 - Portfolio weight

A \$14,800 portfolio contains \$4,440 in stocks.

Weight = Asset value / Portfolio value × 100%

Workings:

Weight = \$4,440 / \$14,800 × 100%

\$4,440 / \$14,800 = 0.30

0.30 × 100% = 30%

Answer: 30% stock weight.

What the answer means: Thirty cents of every portfolio dollar is currently allocated to stocks.

Investment implication: Weights are the language of portfolio construction because they allow different portfolio sizes to be compared.

21.2 Target allocation example

Asset class	Target weight	Dollar target on \$14,800
Stocks	30%	\$4,440
Bonds	25%	\$3,700
Real estate	15%	\$2,220
Broad ETF/other diversified exposure	10%	\$1,480
Commodities/alternatives	5%	\$740
Forex/hedge allocation	5%	\$740
Cash	10%	\$1,480

Target dollars = Portfolio value × Target weight

Example 21.2 - Target dollars

Target bond weight is 25% of a \$14,800 portfolio.

Target bond value = Portfolio value × Target weight

Workings:

25% = 0.25

Target value = \$14,800 × 0.25

Target value = \$3,700

Answer: \$3,700 in bonds.

What the answer means: The percentage policy has been converted into an implementable dollar amount.

Investment implication: This calculation is necessary before comparing current holdings with desired allocation.

21.3 Expected portfolio return

Why the formula is weighted: Each asset contributes to expected portfolio return in proportion to the capital allocated to it. A 20% expected return on a position representing only 5% of the portfolio contributes far less to total expected return than the same return on a 40% position.

Expected does not mean guaranteed: The inputs are forecasts. They may be wrong. The calculation is useful because it makes assumptions explicit and allows alternatives to be compared, not because it predicts the future with certainty.

Consistency rule: Do not mix monthly expected returns with annual expected returns in the same weighted calculation. All asset-return assumptions should use the same horizon and preferably a consistent methodology.

Expected portfolio return = Σ (Asset weight $\times$ Expected asset return)

Weights must sum to 100%, and expected returns should be expressed using the same time horizon and units.

Example 21.3 - Weighted expected return

A simplified portfolio has 50% stocks expected to return 8%, 30% bonds expected to return 4%, and 20% cash expected to return 2%.

$$E(R_p) = \Sigma w_i E(R_i)$$

Workings:

Stocks contribution = $0.50 \times 0.08 = 0.0400 = 4.00\%$

Bonds contribution = $0.30 \times 0.04 = 0.0120 = 1.20\%$

Cash contribution = $0.20 \times 0.02 = 0.0040 = 0.40\%$

Expected portfolio return = $4.00\% + 1.20\% + 0.40\% = 5.60\%$

Answer: 5.60% expected portfolio return.

What the answer means: The portfolio's expected return is the weighted combination of the expected returns of its parts.

Investment implication: Increasing the weight of a high-expected-return asset raises expected return only if its own expected return assumption proves reasonable.

21.4 Rebalancing back to target

Rebalancing trade = Target dollar value - Current dollar value

Example 21.4 - Rebalancing amount

Target bond value is \$3,700, but bonds currently equal \$3,150.

Trade amount = Target value - Current value

Workings:

Trade = $\$3,700 - \$3,150$

Trade = $+\$550$

Answer: Buy/add \$550 of bonds.

What the answer means: The bond position is \$550 below target.

Investment implication: Rebalancing forces the portfolio to reconnect actual holdings with the strategic policy.

21.5 Strategic vs tactical allocation

Strategic allocation	Tactical allocation
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Long-term target structure.	Temporary deviation from target.
Driven by objectives and constraints.	Driven by a specific short/medium-term view.
Changed infrequently.	Can change as opportunities/risks change.
Provides benchmark discipline.	Adds active timing risk.

Transition: Strategic allocation establishes the target architecture. Portal 22 tests whether that architecture is genuinely diversified or merely appears diversified.

Portal 22 - Portfolio Diversification

Learning objective	Portfolio Lab application
Understand correlation, concentration, sector, geographic and currency exposure, and liquidity.	Identify hidden concentrations and diversification gaps.

22.1 Diversification means different sources of risk

The hidden-risk problem: Owning five banks is five securities but may still be one large exposure to credit conditions, interest rates and the domestic economy. Owning several global companies may still create one currency exposure. Diversification therefore requires looking beneath ticker symbols.

Diversification has several dimensions: A useful audit considers issuer, sector, geography, currency, economic sensitivity, liquidity, duration and source of cash flow. The objective is not to eliminate all risk; it is to avoid taking the same risk repeatedly without realizing it.

Diversification is dynamic: Relationships between assets can change. Assets that normally move differently may fall together during a crisis as investors seek liquidity. This is why correlation should be monitored rather than treated as permanent.

A portfolio with many holdings can still be concentrated if they respond to the same economic drivers. Diversification is not counting names; it is examining how exposures overlap.

22.2 Concentration

Concentration weight = Exposure value / Total portfolio value × 100%

Example 22.1 - Sector concentration

Technology-related holdings total \$6,216 in a \$14,800 portfolio.

Sector weight = Sector exposure / Portfolio value × 100%

Workings:

$$\text{Sector weight} = \$6,216 / \$14,800 \times 100\%$$

$$\$6,216 / \$14,800 = 0.42$$

$$0.42 \times 100\% = 42\%$$

Answer: 42% technology-sector exposure.

What the answer means: Although the portfolio may own several securities, 42% is exposed to one broad sector.

Investment implication: The learner should ask whether this concentration is deliberate and whether the risk is adequately compensated.

22.3 Correlation

How to read correlation: Correlation describes direction and strength of linear co-movement, not causation. A correlation of 0.80 does not mean one asset causes the other to move. It means their historical returns tended to move in the same direction relatively strongly over the measured sample.

Why low correlation helps: If two assets do not rise and fall at exactly the same time, losses in one may be partly offset by stability or gains in the other. This interaction can reduce total portfolio volatility without necessarily requiring lower-return assets.

Important warning: Correlation depends on the data frequency, time period and market regime. A number calculated from calm markets can be misleading during stress, so scenario analysis should complement historical correlation.

Correlation ρ_{xy} = Covariance(x,y) / ($\sigma_x \times \sigma_y$)

Correlation ranges from -1 to +1. +1 means returns move perfectly together; 0 means no linear relationship; -1 means perfect opposite movement. Real-world correlations change over time.

Correlation	Interpretation
+0.80 to +1.00	Very strong positive co-movement.
+0.30 to +0.80	Positive relationship.
-0.30 to +0.30	Weak/limited linear relationship.
-0.80 to -0.30	Negative relationship.
-1.00 to -0.80	Very strong inverse movement.

22.4 Two-asset portfolio variance

Breaking the formula into parts: The first term measures Asset 1's weighted variance; the second measures Asset 2's weighted variance; the third is the interaction term. The correlation appears only in the interaction term, which is why the relationship between assets changes portfolio risk.

Why weights are squared: Variance is measured in squared return units, so the portfolio weights enter the individual variance terms as squared weights. The final square root converts variance back into standard deviation, which is expressed in the same return units as the original data.

Interpret before deciding: A lower calculated volatility is useful only if the underlying estimates are credible. Portfolio managers should also ask about drawdown, liquidity, tail risk and whether correlations could rise during the scenario that matters most.

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12}$$

$$\text{Portfolio volatility } \sigma_p = \sqrt{\sigma_p^2}$$

This formula demonstrates why correlation matters. Risk is not simply the weighted average of individual volatilities because the assets interact.

Example 22.2 - Diversification with correlation

Portfolio: 60% Asset A, 40% Asset B. Volatility A = 15%, volatility B = 10%, correlation = 0.20.

$$\text{Variance} = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho$$

$$\text{Volatility} = \sqrt{\text{Variance}}$$

Workings:

Convert percentages: 15%=0.15; 10%=0.10

$$\text{Term 1} = (0.60^2)(0.15^2) = 0.36 \times 0.0225 = 0.00810$$

$$\text{Term 2} = (0.40^2)(0.10^2) = 0.16 \times 0.0100 = 0.00160$$

$$\text{Covariance term} = 2 \times 0.60 \times 0.40 \times 0.15 \times 0.10 \times 0.20 = 0.00144$$

$$\text{Variance} = 0.00810 + 0.00160 + 0.00144 = 0.01114$$

$$\text{Volatility} = \sqrt{0.01114} = 0.10555$$

$$0.10555 \times 100\% = 10.56\%$$

Answer: Approximately 10.56% portfolio volatility.

What the answer means: Because the two assets are not perfectly correlated, combined volatility is lower than a simple concentration in the riskier asset.

Investment implication: Diversification benefit depends on correlation as well as individual asset volatility.

Example 22.3 - Higher correlation comparison

Use the same weights and volatilities, but correlation rises from 0.20 to 0.90.

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho$$

$$\sigma_p = \sqrt{\sigma_p^2}$$

Workings:

$$\text{Term 1} = 0.00810$$

$$\text{Term 2} = 0.00160$$

$$\text{Covariance term} = 2 \times 0.60 \times 0.40 \times 0.15 \times 0.10 \times 0.90 = 0.00648$$

$$\text{Variance} = 0.00810 + 0.00160 + 0.00648 = 0.01618$$

$$\text{Volatility} = \sqrt{0.01618} = 0.12720$$

$$0.12720 \times 100\% = 12.72\%$$

Answer: Approximately 12.72% portfolio volatility.

What the answer means: The portfolio becomes riskier when the assets move more closely together.

Investment implication: Diversification can weaken during crises because correlations among risky assets often rise.

22.5 Geographic and currency exposure

Geographic diversification does not automatically remove currency risk. A domestic investor owning foreign assets may experience a return from the asset itself and a separate return from the exchange rate.

Approximate domestic-currency return $\approx (1 + \text{Local asset return})(1 + \text{Currency return}) - 1$

Example 22.4 - Foreign asset plus currency

A foreign stock rises 8% in its local currency. The foreign currency appreciates 5% against the investor's home currency.

Domestic return = $(1 + \text{Local return})(1 + \text{FX return}) - 1$

Workings:

$$\text{Domestic return} = (1 + 0.08)(1 + 0.05) - 1$$

$$= 1.08 \times 1.05 - 1$$

$$= 1.134 - 1$$

$$= 0.134 = 13.4\%$$

Answer: 13.4% domestic-currency return.

What the answer means: The investor benefited from both the local asset gain and currency appreciation.

Investment implication: A diversified global portfolio should track both asset and currency exposures.

22.6 Liquidity diversification

A portfolio can be diversified by asset class yet still face a liquidity problem. Private businesses, direct property and some alternatives may be difficult to sell quickly. A crisis can make liquidity especially valuable.

Exposure type	Typical liquidity question
Cash/T-bills	Can funds be accessed immediately?
Large listed stocks/ETFs	Can normal-sized trades execute with limited slippage?
Corporate bonds	Is the issue actively traded?
Direct property	How long might a sale take?
Private business	Is there a realistic buyer?
Alternative funds	Are there lockups or redemption limits?

Transition: Diversification identifies where portfolio risk is concentrated. Portal 23 converts that diagnosis into action: add, hold, trim, exit or replace.

Portal 23 - Add, Trim or Exit

Learning objective	Portfolio Lab application
Understand rebalancing bands, valuation discipline, thesis deterioration, sell rules and capital rotation.	For every holding choose ADD, HOLD, TRIM, EXIT or REPLACE.

23.1 Portfolio management is an active set of choices

Holding is a decision: Doing nothing should not mean that the portfolio has been forgotten. HOLD is an active conclusion that the thesis remains intact, valuation remains acceptable and the position size remains appropriate.

Separate thesis from position size: A learner may still like a company but trim it because the position has become too large. Conversely, a falling price does not automatically justify adding; the learner must first establish that the thesis remains valid and the new valuation compensates for risk.

Use pre-committed rules: Written sell rules and rebalancing bands reduce the chance that fear, greed or attachment to a holding will dominate the decision when markets become volatile.

Decision	Meaning	Typical reason
ADD	Increase position.	Thesis strengthens, valuation improves, or position is below target.
HOLD	Keep unchanged.	Thesis intact and size appropriate.
TRIM	Reduce exposure.	Concentration, valuation or risk has risen.
EXIT	Close position.	Thesis broken, risk unacceptable or capital has better use.
REPLACE	Exit one opportunity and fund another.	Relative attractiveness has changed.

23.2 Rebalancing bands

Why use a band instead of an exact target every day? Market prices move continuously. Trading every small deviation would create unnecessary costs and attention. A band allows normal movement while defining the point at which portfolio drift becomes large enough to require review.

Bands are not automatic commands: Crossing a band is a trigger for analysis. Before trading, the learner should check transaction costs, taxes where relevant, liquidity, new information and whether the strategic target itself still makes sense.

A rebalancing band is a tolerance range around a target weight. It avoids unnecessary trading for tiny deviations but creates discipline when exposures move too far.

Upper band = Target weight + Allowed deviation

Lower band = Target weight - Allowed deviation

Example 23.1 - Rebalancing band

Target stock weight is 30% with a ± 5 percentage-point band.

Upper band = Target + Deviation

Lower band = Target - Deviation

Workings:

Upper band = $30\% + 5\% = 35\%$

Lower band = $30\% - 5\% = 25\%$

Answer: Acceptable band = 25% to 35%.

What the answer means: The portfolio does not require a mechanical rebalance while the stock weight remains within the specified range.

Investment implication: Bands create a decision rule before emotion enters the process.

23.3 Calculating how much to trim

Target position value = Current portfolio value × Target weight

Trim amount = Current position value - Target position value

Example 23.2 - Trim to target

Portfolio value is \$16,000. A stock position is worth \$6,000. Target weight is 30%.

Target value = Portfolio × Target weight

Trim = Current value - Target value

Workings:

Target value = \$16,000 × 0.30 = \$4,800

Trim amount = \$6,000 - \$4,800 = \$1,200

Answer: Trim \$1,200.

What the answer means: Selling \$1,200 would return the position to the 30% target, ignoring market movement and transaction costs.

Investment implication: A trim can reduce concentration without rejecting the investment thesis.

23.4 Valuation discipline

An investment thesis can remain correct while the price becomes unattractive. Portfolio management therefore separates the quality of the asset from the attractiveness of the current price.

Upside to estimated value = (Estimated value - Market price) / Market price × 100%

Example 23.3 - Remaining valuation upside

Estimated value is \$72 per share and current price is \$66.

Upside = (Value - Price) / Price × 100%

Workings:

Difference = \$72 - \$66 = \$6

Upside = \$6 / \$66 × 100%

\$6 / \$66 = 0.090909

0.090909 × 100% = 9.09%

Answer: Approximately 9.09% upside to estimated value.

What the answer means: The remaining valuation gap is much smaller than if the shares were bought at a substantially lower price.

Investment implication: If risk has not fallen proportionately, the expected reward-to-risk balance may no longer justify the original position size.

23.5 Capital rotation

Relative expected-return advantage = Expected return of new opportunity - Expected return of current holding

Example 23.4 - Replacement decision

Current holding expected return is 5%. A new opportunity is estimated at 9% with similar risk and liquidity.

Relative advantage = New expected return - Current expected return

Workings:

Relative advantage = 9% - 5%

Relative advantage = 4 percentage points

Answer: 4 percentage points.

What the answer means: The new opportunity has a higher estimated return by 4 percentage points under the assumptions.

Investment implication: Replacement should still account for taxes, costs, uncertainty and thesis quality rather than relying on one forecast.

23.6 Thesis deterioration checklist

- Revenue or cash-flow driver weakens structurally.
- Competitive advantage deteriorates.
- Debt or refinancing risk rises materially.
- Management credibility falls.
- Regulatory change damages economics.
- Valuation no longer compensates for risk.
- Position becomes too concentrated.
- A superior opportunity emerges.
- Original catalyst fails or investment horizon changes.

Transition: Portal 23 creates portfolio rules for action. Portal 24 adds the economic environment, helping the learner understand why several holdings may change together.

Portal 24 - Macroeconomics for Investors

Learning objective	Portfolio Lab application
Understand GDP, inflation, interest rates, unemployment, monetary/fiscal policy, yield curves and economic cycles.	Assess implications across every major portfolio asset class.

24.1 Why macroeconomics matters

Macro works through transmission channels: A rate increase can raise mortgage costs, reduce property affordability, increase corporate borrowing costs, change bond prices and affect currency demand. The same macro event can therefore affect several portfolio holdings through different mechanisms.

Do not turn macro analysis into fortune telling: The purpose is not to predict every recession or policy move. The purpose is to understand sensitivities: what would happen to the portfolio if growth slows, inflation persists or rates stay higher than expected?

Connect macro to cash flows and discount rates: For any macro event, ask two questions. First, does it change the cash flows an asset may generate? Second, does it change the return investors require to hold those cash flows? These two channels explain much of asset-price behavior.

Macroeconomic conditions influence discount rates, borrowing costs, company revenues, currencies, property demand and investor risk appetite. Macro analysis should not replace company or asset analysis; it provides the environment in which those assets operate.

Indicator	Investor question
GDP growth	Is economic activity accelerating or slowing?
Inflation	Are prices rising faster or slower?
Interest rates	How expensive is capital?
Unemployment	How strong is labor demand and household income?
Fiscal policy	Is government spending/tax policy stimulating or restraining activity?
Monetary policy	Is the central bank tightening or easing financial conditions?
Yield curve	What does the term structure imply about rates and economic expectations?

24.2 Real interest rate

Approximate real interest rate $\approx$ Nominal interest rate - Inflation rate

Example 24.1 - Approximate real rate

Nominal policy/market rate is 5.5% and inflation is 3.2%.

Real rate $\approx$ Nominal rate - Inflation

Workings:

Real rate $\approx$ 5.5% - 3.2%

Real rate $\approx$ 2.3%

Answer: Approximately +2.3%.

What the answer means: Borrowers and investors face a positive rate after a simple inflation adjustment.

Investment implication: Higher real rates can tighten financial conditions and reduce the present value of long-duration assets.

24.3 Exact real rate

Exact real rate = $(1 + \text{Nominal rate}) / (1 + \text{Inflation}) - 1$

Example 24.2 - Exact real rate

Nominal rate 5.5%; inflation 3.2%.

Real = $(1+n)/(1+\pi) - 1$

Workings:

Convert to decimals: $n=0.055$; $\pi=0.032$

Real = $1.055 / 1.032 - 1$

$1.055 / 1.032 = 1.0222868$

$1.0222868 - 1 = 0.0222868$

$0.0222868 \times 100\% = 2.22868\%$

Answer: Approximately 2.23%.

What the answer means: The exact Fisher relationship gives a slightly lower result than the simple subtraction approximation.

Investment implication: Use the exact formula when precision matters or inflation/rates are high.

24.4 Yield curve

What the curve contains: Each point represents a yield for a particular maturity. Comparing short and long maturities gives information about current monetary conditions, expected future rates, inflation expectations, term premium and risk appetite.

Why inversion receives attention: When short yields exceed longer yields, markets may be pricing restrictive near-term policy followed by weaker growth or future easing. However, the curve is influenced by many forces, so inversion should be treated as evidence rather than certainty.

Portfolio use: The curve can inform duration choices, bank analysis, refinancing assumptions and the relative attractiveness of short- versus long-maturity bonds.

The yield curve plots bond yields against maturities. A normal upward-sloping curve often shows longer maturities yielding more than short maturities. A flat or inverted curve can reflect expectations of slower growth, future rate cuts or unusually restrictive short-term policy.

Term spread = Long-term government yield - Short-term government yield

Example 24.3 - Yield-curve spread

10-year government yield is 4.0%; 2-year yield is 4.7%.

Term spread = 10Y yield - 2Y yield

Workings:

Term spread = 4.0% - 4.7%

Term spread = -0.7 percentage points

-0.7 percentage points = -70 basis points

Answer: -0.70% or -70 bps.

What the answer means: The selected section of the curve is inverted because the longer yield is below the shorter yield.

Investment implication: An inversion is a signal to investigate economic and policy expectations; it is not a guaranteed recession timer.

24.5 Inflation and different assets

Asset	Potential inflation sensitivity
Cash	Real purchasing power can erode if cash yield lags inflation.
Nominal bonds	Fixed payments lose real value; rising yields can reduce prices.
Stocks	Impact depends on pricing power, input costs and valuation.
Real estate	Rents may rise, but financing costs can also rise.
Commodities	Some can respond strongly to supply shocks/inflation.
Forex	Relative inflation and policy expectations affect currencies.

24.6 Economic cycle map

Phase	Typical features	Portfolio questions
Expansion	Growth strong, employment improving.	Are cyclicals and credit risks being underpriced?
Late cycle	Capacity tight, inflation/rates may rise.	Is leverage becoming dangerous?
Slowdown/recession	Growth weakens, unemployment rises.	Which earnings are cyclical? Which assets

		provide liquidity?
Recovery	Policy easier, activity stabilizes.	Where are valuations still reflecting recession?

Transition: Macroeconomics explains broad forces. Portal 25 focuses on new asset-specific information and teaches the learner to decide whether a headline actually changes the thesis.

Portal 25 - Earnings, News & New Information

Learning objective	Portfolio Lab application
Understand earnings, guidance, analyst revisions, regulation, rate announcements and property-market information.	Distinguish thesis-changing evidence from market noise.

25.1 Information matters relative to expectations

The expectation gap: Asset prices reflect collective expectations before an announcement arrives. Therefore the key comparison is often actual information versus expected information. A result that is objectively strong may still disappoint if the market had priced in something even stronger.

Separate fact, interpretation and price: First identify what was reported. Second determine how it changes the investment thesis. Third observe the market reaction. Starting with the price move can tempt the learner to invent a story after the fact.

Update rather than restart: New information should modify the existing thesis. The learner should ask which assumption changed - revenue growth, margins, financing, competitive position, regulation or valuation - and by how much.

Markets do not respond only to whether news is 'good' or 'bad'. They respond to the difference between new information and what was already expected. A company can report record profit and still fall if investors expected more.

25.2 Earnings surprise

$$\text{Earnings surprise \%} = (\text{Actual EPS} - \text{Expected EPS}) / |\text{Expected EPS}| \times 100\%$$

Example 25.1 - Positive earnings surprise

Actual EPS is \$2.10 while consensus expected \$1.95.

$$\text{Surprise} = (\text{Actual} - \text{Expected}) / |\text{Expected}| \times 100\%$$

Workings:

$$\text{Difference} = \$2.10 - \$1.95 = \$0.15$$

$$\text{Surprise} = \$0.15 / \$1.95 \times 100\%$$

$$\$0.15 / \$1.95 = 0.076923$$

$$0.076923 \times 100\% = 7.69\%$$

Answer: Approximately +7.69%.

What the answer means: Reported EPS exceeded the consensus estimate by about 7.69%.

Investment implication: The investor must still ask whether the beat came from sustainable operations or temporary/accounting factors.

25.3 Guidance revision

$$\text{Guidance midpoint} = (\text{Low guidance} + \text{High guidance}) / 2$$

$$\text{Guidance change \%} = (\text{New midpoint} - \text{Old midpoint}) / \text{Old midpoint} \times 100\%$$

Example 25.2 - Guidance revision

Old revenue guidance was \$980m-\$1.02bn. New guidance is \$1.04bn-\$1.10bn.

$$\text{Midpoint} = (\text{Low} + \text{High}) / 2$$

Change = (New midpoint - Old midpoint)/Old midpoint × 100%

Workings:

Old midpoint = $(\$980\text{m} + \$1,020\text{m}) / 2 = \$1,000\text{m}$

New midpoint = $(\$1,040\text{m} + \$1,100\text{m}) / 2 = \$1,070\text{m}$

Change = $(\$1,070\text{m} - \$1,000\text{m}) / \$1,000\text{m} \times 100\%$

= $\$70\text{m} / \$1,000\text{m} \times 100\% = 7.0\%$

Answer: Guidance midpoint increased 7.0%.

What the answer means: Management raised its central revenue expectation materially.

Investment implication: A thesis based on weak growth may need revision if the guidance improvement is credible and supported by demand evidence.

25.4 Price reaction vs thesis impact

Event	May be noise if...	May change thesis if...
One-day price fall	No new fundamental information.	Reflects new evidence of weaker economics.
Analyst downgrade	Based mainly on short-term target price.	Contains new credible evidence affecting cash flows.
Regulatory headline	Proposal uncertain and distant.	Rule directly changes costs, access or profitability.
Rate announcement	Already fully expected.	Path of rates materially changes financing/valuation.
Property data	One month's volatility.	Vacancy, rents or financing trends change structurally.

25.5 Event return

$$\text{Event return} = (\text{Post-event price} - \text{Pre-event price}) / \text{Pre-event price} \times 100\%$$

Example 25.3 - Price reaction

A stock closes at \$50 before earnings and \$46 after the announcement.

$$\text{Event return} = (P_{\text{after}} - P_{\text{before}}) / P_{\text{before}} \times 100\%$$

Workings:

$$\text{Price change} = \$46 - \$50 = -\$4$$

$$\text{Return} = -\$4 / \$50 \times 100\%$$

$$-\$4 / \$50 = -0.08$$

$$-0.08 \times 100\% = -8.0\%$$

Answer: -8.0% event return.

What the answer means: The stock fell 8% around the event.

Investment implication: The calculation measures reaction, not whether the market is correct. The learner should identify the information that changed expectations.

25.6 Information triage framework

A disciplined sequence: The framework prevents a headline from becoming an immediate trade. Source quality comes first, then expectation, then economic impact, then portfolio action. This slows the decision just enough to distinguish evidence from emotion.

Materiality matters: Not every true fact is investment-relevant. A small operational event may have no meaningful effect on long-term cash flows. The learner should focus on information large enough to change value, risk, liquidity or the probability of the thesis succeeding.

1. What exactly changed?
2. Is the source primary and credible?
3. Was the information already expected?
4. Does it alter cash flows, risk, valuation or financing?
5. Is the change temporary or structural?
6. Does it affect the whole portfolio or one holding?
7. Does the thesis need ADD, HOLD, TRIM, EXIT or no action?

Transition: Portal 25 separates signal from noise. Portal 26 formalizes portfolio risk so that decisions are not made only from narratives or recent price moves.

Portal 26 - Risk Management & Stress Testing

Learning objective	Portfolio Lab application
Understand volatility, beta, drawdown, Sharpe ratio, correlation, VaR, leverage, liquidity and scenario analysis.	Stress-test the portfolio against recession, inflation, property, currency and company shocks.

26.1 Risk is multi-dimensional

Why one risk number is insufficient: Two portfolios can have the same volatility but very different vulnerabilities. One may be liquid and diversified; another may contain leverage, concentrated credit risk or assets that cannot be sold quickly. Risk measurement therefore needs several lenses.

Risk versus uncertainty: Historical statistics summarize what happened in the observed sample. They do not guarantee the future will follow the same distribution. Stress tests deliberately step outside normal historical behavior to ask what happens when assumptions fail.

Portfolio-manager question: Do not ask only, “How much can I make?” Also ask, “How can I lose, how quickly can I lose, what could force me to sell, and can the portfolio survive long enough for the thesis to work?”

No single statistic captures portfolio risk. Volatility measures variability; beta measures market sensitivity; drawdown measures peak-to-trough loss; Sharpe ratio compares excess return with volatility; VaR estimates a loss threshold under assumptions; leverage amplifies exposure; liquidity risk affects the ability to exit.

26.2 Average return

Arithmetic mean return = $\Sigma R_i / n$

Example 26.1 - Average return

Monthly returns are 2%, -1%, 3%, 0%, and 1%.

Mean = $\Sigma R / n$

Workings:

Sum = 2% + (-1%) + 3% + 0% + 1% = 5%

n = 5

Mean = 5% / 5 = 1%

Answer: 1% average monthly return.

What the answer means: The arithmetic average summarizes the central return across observations.

Investment implication: Average return alone does not describe how unstable the path was.

26.3 Sample volatility

Why deviations are squared: Positive and negative deviations would cancel if simply added. Squaring makes all deviations positive and gives greater weight to larger moves. Dividing by n-1 produces the sample variance estimate; taking the square root returns the measure to return units.

What volatility does not say: Volatility treats upside and downside variation symmetrically. Investors may care much more about permanent loss, drawdown or inability to meet cash needs, so volatility should be paired with other risk measures.

Sample variance $s^2 = \Sigma (R_i - \bar{R})^2 / (n - 1)$

Sample volatility $s = \sqrt{s^2}$

Example 26.2 - Sample volatility

Use monthly returns 2%, -1%, 3%, 0%, 1%. The mean is 1%.

$$s^2 = \Sigma(R_i - \bar{R})^2 / (n-1)$$

$$s = \sqrt{s^2}$$

Workings:

Deviations: 1%, -2%, 2%, -1%, 0%

In decimals: 0.01, -0.02, 0.02, -0.01, 0

Squared deviations: 0.0001, 0.0004, 0.0004, 0.0001, 0

Sum of squares = 0.0010

Variance = 0.0010 / (5-1) = 0.00025

Volatility = $\sqrt{0.00025}$ = 0.015811

0.015811 × 100% = 1.58%

Answer: Approximately 1.58% sample monthly volatility.

What the answer means: Monthly returns varied around their 1% mean with a sample standard deviation of about 1.58%.

Investment implication: Higher volatility means a wider historical spread of returns, but does not identify every type of risk.

26.4 Beta

$$\text{Beta } \beta = \text{Covariance}(\text{asset, market}) / \text{Variance}(\text{market})$$

Beta estimates how sensitive an asset's returns have been to market movements. Beta above 1 indicates higher historical market sensitivity; below 1 indicates lower sensitivity; negative beta implies inverse co-movement in the sample.

Example 26.3 - Interpreting beta

A stock has beta 1.30. The market rises 4%. Use the simple beta approximation for the market-driven component.

$$\text{Approximate market-driven stock move} \approx \text{Beta} \times \text{Market move}$$

Workings:

$$\text{Approximate move} = 1.30 \times 4\%$$

$$\text{Approximate move} = 5.2\%$$

Answer: Approximately +5.2% market-related move.

What the answer means: Under a simplified beta relationship, the stock would be expected to move more than the market.

Investment implication: Beta is not a forecast of exact return and does not capture company-specific news.

26.5 Drawdown

Why drawdown feels different from volatility: Drawdown starts from the investor's previous high-water mark and asks how far wealth has fallen. It therefore connects directly to the experience of losing capital and to the difficulty of recovery.

Path dependence: The order of returns matters to the investor. A portfolio that falls deeply early may require a large percentage recovery before compounding can resume from the old peak. This is why protecting against catastrophic loss can be more important than maximizing every short-run gain.

$$\text{Drawdown} = (\text{Current or trough value} - \text{Prior peak value}) / \text{Prior peak value} \times 100\%$$

Example 26.4 - Maximum drawdown episode

A portfolio peaks at \$18,000 and later falls to \$14,400 before recovering.

$$\text{Drawdown} = (\text{Trough} - \text{Peak}) / \text{Peak} \times 100\%$$

Workings:

$$\text{Loss} = \$14,400 - \$18,000 = -\$3,600$$

$$\text{Drawdown} = -\$3,600 / \$18,000 \times 100\%$$

$$-\$3,600 / \$18,000 = -0.20$$

$$-0.20 \times 100\% = -20\%$$

Answer: -20% drawdown.

What the answer means: The portfolio lost 20% from its prior peak to the trough.

Investment implication: Drawdown is intuitive for investors because it measures the decline they actually had to endure.

26.6 Recovery mathematics

$$\text{Required recovery return} = (\text{Peak value} / \text{Trough value}) - 1$$

Example 26.5 - Recovery after a 20% drawdown

Portfolio falls from \$18,000 to \$14,400. What return is required to get back to \$18,000?

Recovery = Peak / Trough - 1

Workings:

Recovery = $\$18,000 / \$14,400 - 1$

$\$18,000 / \$14,400 = 1.25$

$1.25 - 1 = 0.25$

$0.25 \times 100\% = 25\%$

Answer: 25% recovery return required.

What the answer means: A 20% loss requires a 25% subsequent gain to recover because the recovery starts from a smaller capital base.

Investment implication: Avoiding large drawdowns can be important even when long-run expected return is attractive.

26.7 Sharpe ratio

What is being compared: The numerator is the return earned above a risk-free alternative. The denominator is portfolio volatility. The ratio therefore asks how much excess return was obtained for each unit of measured variability.

Comparison rule: Sharpe ratios should be compared only when the return horizon, risk-free rate, data frequency and volatility methodology are consistent. A ratio calculated from monthly data cannot be casually compared with one calculated differently.

Limitation: The Sharpe ratio compresses risk into volatility and can be misleading for investments with asymmetric or infrequent losses. It should support, not replace, drawdown and scenario analysis.

Sharpe ratio = (Portfolio return - Risk-free return) / Portfolio volatility

Example 26.6 - Sharpe ratio

Portfolio return 9%; risk-free return 3%; portfolio volatility 12%.

Sharpe = $(R_p - R_f) / \sigma_p$

Workings:

Excess return = 9% - 3% = 6%

Convert consistently: 0.06 / 0.12

Sharpe = 0.50

Answer: Sharpe ratio = 0.50.

What the answer means: The portfolio earned 0.50 units of excess return per unit of volatility under this simplified historical/expected framework.

Investment implication: Sharpe ratios are most meaningful when calculated consistently over comparable periods and return distributions.

26.8 Value at Risk - simple parametric illustration

How to read VaR correctly: A 95% one-day VaR of \$1,974 does not mean the maximum one-day loss is \$1,974. It means the model places the threshold near the boundary of the worst 5% of modeled daily outcomes. Losses beyond the threshold can be much larger.

Why assumptions matter: The normal parametric method assumes a particular return distribution and stable volatility. During crises, volatility and correlations can jump and returns can have fatter tails than the normal distribution. This is why VaR must be paired with stress testing.

VaR estimates a loss threshold that should not be exceeded with a stated confidence over a stated horizon, under model assumptions. It does not estimate the worst possible loss.

One-period normal VaR $\approx$ Portfolio value $\times (z \times \sigma - \mu)$

For a simple zero-mean illustration, $\text{VaR} \approx \text{Portfolio value} \times z \times \sigma$.

Example 26.7 - 95% one-period VaR illustration

Portfolio value \$100,000; one-day volatility 1.2%; assume zero mean and $z=1.645$.

$\text{VaR} \approx V \times z \times \sigma$

Workings:

$\sigma = 1.2\% = 0.012$

$\text{VaR} \approx \$100,000 \times 1.645 \times 0.012$

$1.645 \times 0.012 = 0.01974$

$\$100,000 \times 0.01974 = \$1,974$

Answer: Approximately \$1,974 at 95% one-day VaR.

What the answer means: Under the normal-model assumptions, there is a 5% tail beyond a loss threshold of about \$1,974 over one day.

Investment implication: VaR can underestimate extreme events and depends heavily on volatility, distribution and horizon assumptions.

26.9 Leverage

Leverage multiple = Gross exposure / Investor equity

Example 26.8 - Portfolio leverage

Investor equity is \$20,000 and gross market exposure is \$30,000.

Leverage = Gross exposure / Equity

Workings:

Leverage = \$30,000/\$20,000

Leverage = 1.5×

Answer: 1.5× leverage.

What the answer means: The investor controls \$1.50 of gross exposure for every \$1 of equity.

Investment implication: Leverage magnifies both gains and losses and can create forced selling if financing terms tighten.

26.10 Scenario stress testing

Stress testing reverses the question: Instead of asking what loss is likely under a statistical distribution, scenario analysis asks what the portfolio would experience if a specified event occurred. This makes it especially useful for risks that are rare, structural or difficult to model from limited historical data.

Design meaningful shocks: A good stress scenario links shocks logically. A recession scenario might combine lower equity prices, wider credit spreads, weaker property values and perhaps lower government yields. Random independent shocks can miss how real crises propagate through a portfolio.

Use the result to change decisions: If a plausible scenario produces a loss larger than the investor can tolerate, the answer is not merely to record the number. Position size, leverage, liquidity, hedges or asset allocation may need to change.

Scenario analysis applies shocks to portfolio exposures. It does not require the shock probabilities to be known precisely; its purpose is to reveal vulnerability.

Estimated scenario P/L = Σ (Position value $\times$ Assumed scenario return)

Example 26.9 - Multi-asset recession stress

Portfolio: \$6,000 stocks (-20%), \$4,000 bonds (+5%), \$3,000 real estate (-15%), \$1,800 cash (0%).

Scenario P/L = $\Sigma \text{Value}_i \times \text{Shock return}_i$

Workings:

Stocks P/L = \$6,000 $\times$ (-0.20) = -\$1,200

Bonds P/L = \$4,000 $\times$ 0.05 = +\$200

Real estate P/L = \$3,000 $\times$ (-0.15) = -\$450

Cash P/L = \$1,800 $\times$ 0 = \$0

Total P/L = -\$1,200 + \$200 - \$450 + \$0 = -\$1,450

Answer: -\$1,450 scenario loss.

What the answer means: The assumed recession shock would reduce the portfolio by \$1,450 under the stated simplified shocks.

Investment implication: The stress identifies which exposures drive loss and whether bonds/cash provide enough protection.

Scenario return = Scenario P/L / Starting portfolio value × 100%

Example 26.10 - Scenario return

Using the prior stress test, starting portfolio value is \$14,800 and scenario P/L is -\$1,450.

Scenario return = P/L / Starting value × 100%

Workings:

Scenario return = $-\$1,450 / \$14,800 \times 100\%$

$-\$1,450 / \$14,800 = -0.097973$

$-0.097973 \times 100\% = -9.80\%$

Answer: Approximately -9.80%.

What the answer means: The simplified recession scenario would reduce portfolio value by about 9.8%.

Investment implication: The learner can compare this loss with risk tolerance, liquidity needs and the mandate.

26.11 Liquidity stress

Risk models often assume assets can be traded. In practice, a crisis may widen spreads or make some assets difficult to sell. Stress testing should therefore include a liquidity question: 'What if I cannot sell this position at the displayed price?'

Stress	Questions
Recession	Which revenues, credit spreads and property values fall?
Inflation shock	Which bonds lose value? Which firms can pass through costs?
Rate shock	How do duration, mortgage costs and equity valuations respond?
Currency shock	Which foreign exposures gain or lose?
Company shock	What if the largest holding falls 40%?
Liquidity shock	What if property/private assets cannot be sold for months?

Transition: Portal 26 quantifies vulnerabilities. Portal 27 removes the comfort of isolated exercises and asks the learner to make portfolio decisions while several risks occur at once.

Portal 27 - Green City Crisis Challenge

Learning objective	Portfolio Lab application
Integrate a market-shock simulation across multiple holdings and asset classes.	Decide what to sell, protect, add to or leave untouched under pressure.

27.1 The crisis challenge is a decision test

Process becomes more valuable under pressure: During a crisis, prices move faster than a learner can fully research every development. A pre-existing mandate, liquidity plan, rebalancing policy and thesis journal reduce the temptation to improvise every decision.

Falling price is not the same as rising risk: Sometimes a price falls while long-term fundamentals remain intact, improving expected return. In other cases the price falls because the asset's economics have genuinely deteriorated. The learner must identify which case applies before acting.

Cash has option value: Liquidity allows the portfolio to meet obligations without forced selling and can provide capital to buy attractive assets during dislocation. Holding some cash can therefore be a strategic choice rather than evidence of indecision.

A crisis creates incomplete information, fast price changes and emotional pressure. The learner should not react to every falling price. The goal is to separate liquidity needs, thesis breaks, valuation opportunities and portfolio-level risk.

27.2 Crisis triage order

Why liquidity is first: An investor who must raise cash immediately has fewer choices. Checking obligations, leverage and financing before valuation prevents the learner from designing an elegant long-term response that cannot actually be implemented.

Why thesis breaks come before bargains: A 30% price decline does not automatically create a bargain. If expected cash flows have deteriorated by more than the price, the asset may actually be less attractive. Updated fundamentals must come before averaging down.

8. Check liquidity: do any near-term obligations require cash?
9. Identify forced risks: margin calls, refinancing, covenant or redemption pressure.
10. Identify thesis breaks: which assumptions are no longer valid?
11. Measure concentration: which positions now dominate portfolio risk?
12. Estimate scenario losses and recovery capacity.
13. Compare valuation with updated fundamentals.
14. Decide ADD, HOLD, TRIM, EXIT, REPLACE or HEDGE for each holding.
15. Record the reason before acting.

27.3 Crisis scenario

Integrated shock

Assume: policy rates rise unexpectedly by 1.0 percentage point; equity markets fall 15%; a portfolio property falls 10%; corporate credit spreads widen; the home currency appreciates 6%; and one major company holding cuts guidance.

27.4 Position-by-position response grid

Holding	New information	Portfolio question	Possible action
Long-duration bond	Rates +1%.	How much duration loss is	Trim/Hold depending on

		expected?	mandate.
High-quality short bond	Credit stable.	Does it provide liquidity/stability?	Hold/Add.
Growth stock	Market down + rates up.	Has value fallen more than fundamentals?	Hold/Trim/Add after valuation.
Weak company	Guidance cut.	Has thesis broken?	Trim/Exit.
Direct property	Value -10%; financing cost rising.	Can debt be serviced?	Hold/Protect/Exit if leverage unsafe.
Foreign asset	Home currency +6%.	Is FX loss hedged?	Hedge/Hold.
Cash	Stable nominal value.	Does it preserve optionality?	Hold/deploy selectively.

27.5 Crisis duration calculation

Example 27.1 - Rate shock on bond allocation

A \$3,700 bond allocation has modified duration 5.0. Yield rises 1.0 percentage point.

Approximate $\% \Delta \text{Price} \approx -\text{Duration} \times \Delta \text{Yield}$

Estimated dollar P/L $\approx \text{Position value} \times \% \Delta \text{Price}$

Workings:

$\Delta \text{Yield} = 1.0\% = 0.010$

$\% \Delta \text{Price} \approx -5.0 \times 0.010 = -0.050 = -5.0\%$

Dollar P/L $\approx \$3,700 \times (-0.05)$

Dollar P/L $\approx -\$185$

Answer: Approximately $-\$185$.

What the answer means: The duration-based rate shock is estimated to reduce the bond allocation by about \$185, ignoring convexity and credit effects.

Investment implication: The learner can decide whether the loss is acceptable given the stabilizing role of bonds and future reinvestment yields.

27.6 Currency shock calculation

Example 27.2 - Foreign exposure after home-currency appreciation

A foreign asset has 0% local return, but the foreign currency falls 6% against the investor's home currency.

Domestic return $= (1 + \text{Local return})(1 + \text{FX return}) - 1$

Workings:

Local return = 0.00

FX return = -0.06

Domestic return $= (1+0)(1-0.06)-1$

$= 1 \times 0.94 - 1$

$= -0.06 = -6\%$

Answer: -6.0% domestic-currency return.

What the answer means: Even though the underlying asset price did not change locally, currency translation creates a 6% loss for the home-currency investor.

Investment implication: Currency risk can be material even when asset-specific fundamentals are unchanged.

27.7 Concentration after market moves

New weight $= \text{New position value} / \text{New total portfolio value} \times 100\%$

Example 27.3 - Weight changes during crisis

A defensive holding remains \$2,500 while the total portfolio falls from \$14,800 to \$12,500.

New weight $= \text{Position value} / \text{New portfolio value} \times 100\%$

Workings:

New weight $= \$2,500 / \$12,500 \times 100\%$

$\$2,500 / \$12,500 = 0.20$

$$0.20 \times 100\% = 20\%$$

Answer: 20% new portfolio weight.

What the answer means: A position can become a larger percentage of the portfolio even if its own value does not rise.

Investment implication: Crisis rebalancing should use updated weights rather than old target percentages.

27.8 Crisis portfolio decision record

Holding	Action	Evidence changed?	Valuation changed?	Risk contribution	Liquidity impact	Review trigger

27.9 Crisis debrief

- Which losses came from expected market risk and which came from thesis failure?
- Which assets diversified the portfolio when needed?
- Which correlations increased unexpectedly?
- Was liquidity sufficient?
- Did leverage force action?
- Were any positions sold simply because prices fell?
- Were attractive opportunities ignored because cash was unavailable?
- What portfolio rule should change before the next crisis?

Green City completion standard

The learner is ready for Yellow City when they can manage a portfolio as a changing system: measuring exposures, interpreting new information, rebalancing deliberately, stress-testing risk and acting under pressure without abandoning the investment process.

Green City Integrated Calculation Lab

How to use this lab: Do each problem without looking at the worked answer. Write the formula first, label the units, substitute the numbers, calculate intermediate steps and then write one sentence interpreting the result. The interpretation is part of the calculation because portfolio management requires converting numbers into decisions.

Common error check: Percentage points and decimal changes are not interchangeable inside formulas. For example, a 0.50 percentage-point yield change must be entered as 0.005 in a duration multiplication. Always convert deliberately before calculating.

The following exercises reinforce the formulas used throughout Green City. Each demonstrates the required sequence: formula, substitution, workings, answer and interpretation.

Lab A - Bonds

Practice 1 - Current yield

A bond pays \$45 annually and trades at \$900.

Current yield = Coupon / Price × 100%

Workings:

Current yield = $\$45 / \$900 \times 100\%$

= $0.05 \times 100\%$

= 5.0%

Answer: 5.0%.

What the answer means: Coupon income equals 5% of the current purchase price.

Practice 2 - Duration shock

Duration 4.4; yield rises 0.50%.

$\% \Delta P \approx -D \times \Delta y$

Workings:

$\Delta y = 0.005$

$\% \Delta P \approx -4.4 \times 0.005 = -0.022$

= -2.2%

Answer: -2.2%.

What the answer means: The price is estimated to fall about 2.2%.

Lab B - Allocation & Diversification

Practice 3 - Target dollars

Portfolio \$20,000; target real-estate weight 12%.

Target value = Portfolio × Weight

Workings:

Target = $\$20,000 \times 0.12$

= \$2,400

Answer: \$2,400.

What the answer means: The target percentage becomes an implementable dollar allocation.

Practice 4 - Concentration

Three bank stocks total \$5,250 in a \$15,000 portfolio.

Sector weight = Sector value / Portfolio value × 100%

Workings:

$$=\$5,250/\$15,000 \times 100\%$$

$$=0.35 \times 100\%$$

$$=35\%$$

Answer: 35%.

What the answer means: Different stock names can still represent one concentrated sector bet.

Lab C - Rebalancing & Macro

Practice 5 - Trim amount

Portfolio \$25,000; current equity holding \$10,000; target weight 32%.

Target value = Portfolio × Target weight

Trim = Current - Target

Workings:

Target = \$25,000 × 0.32 = \$8,000

Trim = \$10,000 - \$8,000 = \$2,000

Answer: Trim \$2,000.

What the answer means: The trade restores the holding to target.

Practice 6 - Exact real rate

Nominal rate 7%; inflation 4%.

Real = $(1+n)/(1+\pi) - 1$

Workings:

= 1.07 / 1.04 - 1

= 1.028846 - 1

= 0.028846 = 2.8846%

Answer: Approximately 2.88%.

What the answer means: The precise real return/rate is below the simple 3% subtraction approximation.

Lab D - Risk

Practice 7 - Drawdown

Portfolio peak \$50,000; trough \$39,000.

Drawdown = $(\text{Trough} - \text{Peak}) / \text{Peak} \times 100\%$

Workings:

= (\$39,000 - \$50,000) / \$50,000 × 100%

= -\$11,000 / \$50,000 × 100%

= -22%

Answer: -22%.

What the answer means: The portfolio lost 22% from peak to trough.

Practice 8 - Required recovery

Recover from \$39,000 to \$50,000.

Recovery = $\text{Peak} / \text{Trough} - 1$

Workings:

= \$50,000 / \$39,000 - 1

= 1.282051 - 1

= 0.282051 = 28.21%

Answer: Approximately 28.21%.

What the answer means: The gain required after a loss is larger than the original percentage loss.

Practice 9 - Sharpe

Return 11%; risk-free 3%; volatility 16%.

$$\text{Sharpe} = (R_p - R_f) / \sigma_p$$

Workings:

$$\text{Excess return} = 11\% - 3\% = 8\%$$

$$\text{Sharpe} = 0.08 / 0.16$$

$$= 0.50$$

Answer: 0.50.

What the answer means: The portfolio produced 0.5 units of excess return per unit of volatility under the chosen inputs.

Green City Portfolio Management Templates

Strategic allocation sheet

Asset class	Current \$	Current %	Target %	Target \$	Trade required
Stocks					
Bonds					
Real estate					
ETFs/pooled					
Commodities/alternatives					
Forex/hedges					
Cash					

Diversification audit

Exposure	Current weight	Maximum desired	Hidden overlap?	Action
Largest single holding				
Largest sector				
Largest geography				
Largest currency				
Illiquid assets				
Leveraged exposure				

Holding decision sheet

Holding	ADD/HOLD/TRIM/EXIT/REPLACE	Thesis status	Valuation	Risk	Target weight	Next trigger

Macroeconomic Dashboard Template

Indicator	Current observation	Direction	Why it matters	Portfolio implication
GDP growth				
Inflation				
Policy rate				
Unemployment				
10Y-2Y spread				
Credit spreads				
Currency				
Property conditions				

New Information Triage

Event	Expected?	Primary source?	Cash-flow impact	Risk impact	Valuation impact	Action

Stress Test Matrix

Holding	Recession	Inflation	Rates +1%	FX shock	Company shock	Liquidity shock
Stocks						
Bonds						
Real estate						
Forex						
Alternatives						
Cash						

Green City Formula Reference

Using the reference correctly: The formula table is a revision tool, not a substitute for understanding. Before selecting a formula, identify the question being asked, the units available and the assumptions behind the measure. After calculating, always translate the result back into a portfolio-management statement.

Concept	Formula
Bond coupon	Face value × Coupon rate
Current yield	Annual coupon / Market price × 100%
Bond HPR	(Coupon + Ending price - Beginning price)/Beginning price × 100%
Duration estimate	%ΔPrice ≈ -Modified duration × ΔYield
Credit spread	Corporate yield - Comparable government yield
ETF fee	Investment × Expense ratio
Price return	(Ending price - Beginning price)/Beginning price × 100%
Asset weight	Asset value / Portfolio value × 100%
Expected portfolio return	$\sum w_i E(R_i)$
Target dollars	Portfolio value × Target weight
Rebalancing trade	Target value - Current value
Correlation	Covariance/(σ _x σ _y)
Two-asset variance	$w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho$
Domestic foreign-asset return	(1+Local return)(1+FX return)-1
Rebalancing band	Target ± permitted deviation
Valuation upside	(Value-Price)/Price × 100%
Approx. real rate	Nominal rate - Inflation
Exact real rate	(1+Nominal)/(1+Inflation)-1
Term spread	Long yield - Short yield
Earnings surprise	(Actual EPS-Expected EPS)/ Expected EPS × 100%
Event return	(Post-event price-Pre-event price)/Pre-event price × 100%
Mean return	$\sum R/n$
Sample variance	$\sum (R-\bar{R})^2/(n-1)$
Volatility	$\sqrt{\text{Variance}}$
Beta	Cov(asset,market)/Var(market)
Drawdown	(Trough-Peak)/Peak × 100%
Required recovery	Peak/Trough -1
Sharpe ratio	(R _p -R _f)/σ _p
VaR illustration	Portfolio value × z × σ
Leverage	Gross exposure / Equity
Scenario P/L	$\sum (\text{Position value} \times \text{Scenario return})$

Green City Vocabulary Reference

Term	Meaning
Bond	Debt security issued by a government or company.
Coupon	Contractual bond interest payment.
Yield	Return measure implied by price and cash flows.
Duration	Interest-rate sensitivity measure.
Credit spread	Extra yield relative to a benchmark government bond.
ETF	Exchange-traded pooled investment vehicle.
Tracking difference	ETF return minus benchmark return.
Strategic allocation	Long-term target portfolio structure.
Rebalancing	Trading to restore or change target exposures.
Correlation	Measure of linear co-movement between returns.
Concentration	Large exposure to one holding, sector, geography or risk factor.
Drawdown	Peak-to-trough decline.
Beta	Historical market sensitivity.
Sharpe ratio	Excess return per unit of volatility.
VaR	Model-based loss threshold at a chosen probability/horizon.
Stress test	Application of hypothetical shocks to portfolio exposures.
Liquidity risk	Risk of being unable to transact quickly at a reasonable price.
Capital rotation	Moving capital from less attractive to more attractive opportunities.
Thesis deterioration	Evidence that weakens the original investment rationale.
Yield curve	Relationship between yields and maturities.

Green City Completion Challenge

What strong performance looks like: A strong submission does more than produce correct arithmetic. It connects each calculation to evidence and to the persistent portfolio. The learner should be able to explain why a measure was chosen, what assumptions it uses, what the answer means, and what action - if any - follows.

Decision quality over lucky outcomes: A portfolio can rise for the wrong reasons and fall despite a sound process. The Green City standard therefore emphasizes defensible research, sizing, diversification, risk control and willingness to revise a thesis when evidence changes.

The Green City assessment should demonstrate that the learner can manage the portfolio after the initial investment decisions have already been made.

Required work

- Evaluate at least one government or corporate bond and explain coupon, yield, duration and credit risk.
- Compare a direct asset with an ETF or pooled alternative.
- Create a target multi-asset strategic allocation for the persistent portfolio.
- Calculate current and target weights and show any required rebalancing trades.
- Perform a concentration and correlation review.
- Make ADD, HOLD, TRIM, EXIT or REPLACE decisions for every major holding.
- Prepare a macroeconomic dashboard and explain implications across several asset classes.
- Evaluate at least three pieces of new information and distinguish signal from noise.
- Calculate volatility, drawdown, Sharpe ratio and at least one additional risk measure.
- Run at least three stress scenarios, including one multi-asset crisis.
- Defend crisis actions using evidence, formulas, liquidity needs and the portfolio mandate.

Green City readiness checklist

- I can calculate bond coupon, current yield and holding-period return.
- I understand why yields and bond prices move inversely.
- I can use duration to approximate interest-rate sensitivity.
- I can evaluate ETF costs, holdings and tracking.
- I can convert target asset weights into dollar allocations.
- I can calculate expected portfolio return.
- I can identify hidden sector, geographic, currency and liquidity concentration.
- I understand how correlation affects portfolio risk.
- I can calculate rebalancing amounts and apply rebalancing bands.
- I can explain macroeconomic implications for bonds, stocks, property, commodities and currencies.
- I can distinguish thesis-changing information from price noise.
- I can calculate volatility, beta-related sensitivity, drawdown, Sharpe ratio and simple VaR.
- I can run scenario stress tests and interpret losses.
- I can make portfolio decisions under a crisis without reacting blindly to falling prices.

Transition to Yellow City

Green City answers: How do I manage and adapt the portfolio as conditions change? Yellow City will require the learner to formalize investment theses, confront behavioral biases, optimize portfolios, rotate capital, manage historical simulations, invest globally and operate under a professional fund mandate.